

12.1 The Forward Process

Add a little Gaussian noise for T steps until the sample is almost prior noise.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A **diffusion** model learns to denoise. The forward process is not learned. You add Gaussian noise on a schedule until the sample is almost prior noise. Training later asks a net to predict that noise.

1. Destroy the sample on purpose

Start from data x_0 . A variance schedule $0 < \beta_t < 1$ (small at first, larger later) defines

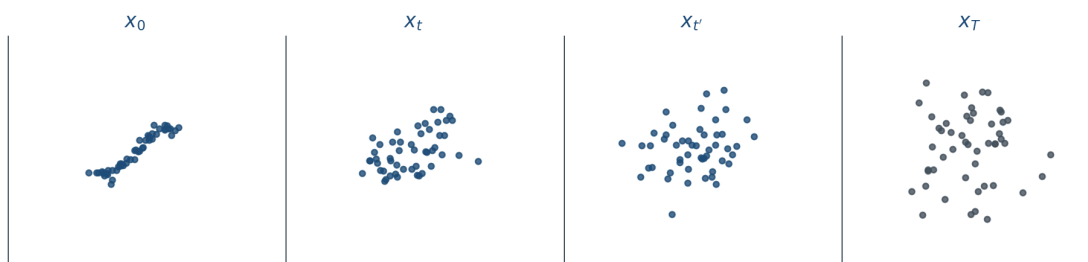
$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

The Markov step is $q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t \mathbf{I})$. You do not have to simulate every hop. The marginal is closed form:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}).$$

In code: $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$.

Forward process: data becomes noise



2. What the schedule is doing

$\bar{\alpha}_t$ starts near 1 (the point is still itself) and falls toward 0 (the point is almost ε). Linear, cosine, and learned β_t are all ways to spend more steps where structure dies. Too fast a schedule and early t is already junk; too slow and you waste capacity on near-copies of x_0 .

T (often 1000 in image papers) is a discretization. Lab 12 uses a small T in 2-D so you can plot the path.

3. Why this is nicer than a GAN to train

The forward kernel is **fixed**. There is no discriminator to race. The training signal at each t is a denoising regression against a known ε . You still have to pick $\{\beta_t\}$ and T . Sampling will be a reverse chain (note **12.2**), which is slower than one GAN forward pass.

4. Practice

1. If $\bar{\alpha}_t = 1$, what is x_t ? If $\bar{\alpha}_t = 0$, what is x_t ?
2. Why can you sample x_t in one line instead of looping t steps of $q(x_s | x_{s-1})$?
3. A classmate sets every $\beta_t = 0.9$. What goes wrong in the first few t ?